

# Richard Álvarez

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## Research Areas

Human–computer interaction; human-centered AI; computational media; AI-assisted creative tools.

## Education

B.A. in Film Aug 2020–May 2024  
Kenyon College Gambier, Ohio  
*Minor in History; Integrated Program in Humane Studies.*

## Publications and Posters

- [1] Álvarez, Richard Anthony. *A Retrieval-Augmented Film Recommendation System*. Poster, *IPHS 484: Senior Seminar*, Paper 34, Digital Kenyon, 2024. Digital Kenyon.
- [2] Álvarez, Richard Anthony. *Captivating Video Editing Secrets: Unsupervised Deep Learning and PySceneDetect Analysis of Creative Strategies*. *IPHS 484: Senior Seminar*, Paper 22, Digital Kenyon, 2023. Digital Kenyon.

## Professional Experience

AI Solutions Engineer Sep 2025–Present  
*Layer 7 Systems, Independent Contractor* Remote

Built and operated a stateful TypeScript, Bun, and Express MCP service connecting Retell voice agents with Autotask and Microsoft Graph/Teams.

Implemented identity and company resolution, contact-safe ticket access, authorization gates, ticket actions, technician routing, post-call processing, and human handoff.

Developed an authenticated React operations dashboard, maintained a 756-test regression suite, and deployed Docker services through GitHub Actions.

Analyzed 60 workflow recordings from 14 users and identified 176 moments suitable for AI assistance.

Built a Next.js and Neon/PostgreSQL learning platform for 34 trainees across four courses and 16 modules.

Freelance Software and Web Developer Oct 2024–Present  
*Richard Álvarez, Sole Proprietorship* Chicago, Illinois

Design and deploy responsive applications using Next.js, React, Node.js, Tailwind CSS, Stripe, PostgreSQL, authentication, and object storage.

Scope, document, deploy, and maintain client applications, hosting, and content workflows.

IT Assistant Sep 2023–Feb 2024  
*Kenyon College Library and Information Services* Gambier, Ohio

Supported workstation imaging, hardware deployment, classroom technology, and recurring maintenance scheduling.

## Selected Software and AI Projects

|                                        |           |
|----------------------------------------|-----------|
| Vigil                                  | 2025–2026 |
| Half-Time Draw Predictor               | 2026      |
| Chroma                                 | 2026      |
| GREasyVocab                            | 2024      |
| Bitset Game of Life                    | 2024–2026 |
| Kubrick-Style Script Generation        | 2024      |
| Rotten Tomatoes Sentiment Analysis     | 2024      |
| DV Tools                               | 2025      |
| Business and AI Training Platform      | 2026      |
| Workflow Capture and Analysis Pipeline | 2026      |
| Configurable Voice-Reception Template  | 2025      |
| Message-Box Auto-Attendant             | 2025      |
| VibeVoice Evaluation Prototype         | 2025      |

## Selected Professional Projects

|                                          |              |
|------------------------------------------|--------------|
| Joaquin Morales Cinematography Portfolio | 2024–2026    |
| Machine Television                       | 2024–2025    |
| BDPA HSCC Website Prototype              | 2025         |
| Personal Portfolio                       | 2024–Present |

## Professional and Community Service

|                                                                                                                                      |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| High School Computer Competition Judge<br><i>BDPA</i>                                                                                | 2024–2025<br><i>Chicago, Illinois</i> |
| Reviewed and scored student programming submissions for the 2024 and 2025 national competitions and completed formal judge training. |                                       |

## Professional Certification

|                                |      |
|--------------------------------|------|
| CompTIA IT Fundamentals (ITF+) | 2024 |
|--------------------------------|------|

## Technical Skills

*Programming languages:* TypeScript, Python, JavaScript, SQL, C++ , HTML/CSS, Shell.

*Frameworks and runtimes:* Bun, Node.js, Express, Hono, FastAPI, React, Next.js, Electron, Zod.

*Data and applied AI:* PostgreSQL, Neon, SQLite FTS5, MCP, REST APIs, tool calling, structured outputs, retrieval-augmented generation, context engineering, OpenAI, Anthropic, Gemini, Retell.

*Enterprise integration:* Autotask, Microsoft Graph, Microsoft Teams.

*Infrastructure and reliability:* Docker, GitHub Actions, Linux/VPS operations, CI/CD, automated testing, validation, structured logging, observability, incident response.

*Computational media:* PySceneDetect, FFmpeg, NLP and sentiment analysis, computational film analysis, interactive media.